

Vincent Thibeault

Wu Tsai & CPRA Postdoctoral Fellow

Yale University

New Haven, CT



Personal website



[first].[last].1@ulaval.ca



VinceThi



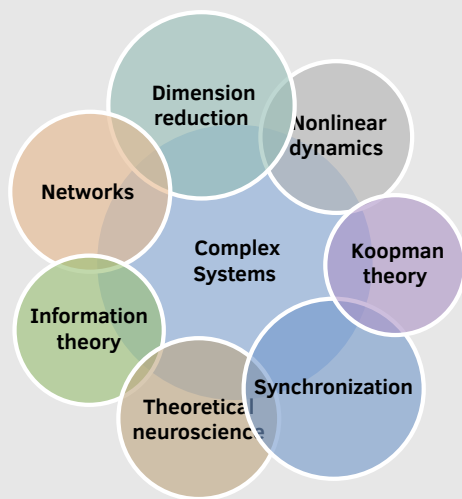
Google scholar



ResearchGate

Overview

Research fields



Programming & Visual Tools

- Python
- Mathematica
- Matlab
- Inkscape

Languages

- French (*Native*)
- English (*Advanced*)
- Spanish (*Intermediate*)

Education

2021 -
2026

Ph.D., Physics

Supervisors: Patrick Desrosiers and Antoine Allard
Board of Honor of the Faculty of Graduate and Postdoctoral Studies

Université Laval

2017 -
2020

M.Sc., Physics

Supervisors: Patrick Desrosiers and Louis J. Dubé
Board of Honor of the Faculty of Graduate and Postdoctoral Studies

Université Laval

2014 -
2017

B.Sc., Physics

Université Laval

Affiliations

2026 -
2029

Wu Tsai Institute

Yale University

2015 -
2026

Physics department

Université Laval

2017 -
2026

CIMMUL (Interdisciplinary Center for Math. Modeling)

2021 -
2026

UNIQUE (Unifying Neuroscience and Artificial Intelligence)

2017 -
2023

Sentinelle Nord

Work experience

2017 -
2022

Graduate Teaching Assistant in Physics

Université Laval

- **Role:** Classroom problem solving, corrector and helper
- **Courses:** [Nonlinear dynamics](#), [Chaos and Complexity](#) (2019, 2022), [Statistical Physics](#) (2019-2021), [Mathematical Physics I](#) (2017, 2018)

2019

Institutional data and network visualization analyst

Université Laval

- **Role:** Analyze and visualize collaborator networks for the [INQ](#).
- **Tools:** Python, D3.js

2016

Undergraduate Research Assistant in Neuroscience

CERVO

- **Role:** Creation of a spatial CA3 neuron model with evoked and spontaneous neurotransmission
- **Keywords:** Hodgkin-Huxley model, miniature, calcium dynamics
- **Tools:** Python, [NEURON](#), [ModelDB](#)
- **Supervisor:** Patrick Desrosiers, Simon Hardy

2015 -
2016

Undergraduate Tutor in Mathematical Physics

Université Laval

- **Role:** Help undergraduate students in the first three courses of mathematical physics
- **Content** From mathematical logic to orthogonal polynomials

2015 -
2016

Undergraduate Research Assistant in Attosecond Physics

COPL

- **Role:** Molecular tomography data analysis and construction of an ion detector
- **Keywords:** Attosecond physics, electronic recollision, FFT
- **Tools:** Matlab, Labview
- **Supervisor:** Julien Beaudoin-Bertrand

Scientific contribution

Refereed Publications

- V. Thibeault, A. Allard, P. Desrosiers, “[The low-rank hypothesis of complex systems](#)”, *Nat. Phys.*, **20**, 294 (2024).
- ★ [Featured in Nature Physics' News & Views.](#)
 - ★ [Cover page of Nature Physics.](#)
- C. Murphy, V. Thibeault, A. Allard, P. Desrosiers, “[Duality between predictability and reconstructability in complex systems](#)”, *Nat. Commun.*, **15**, 4478 (2024).
- M. Vugué, V. Thibeault, P. Desrosiers, A. Allard, “[Dimension reduction of dynamics on modular and heterogeneous directed networks](#)”, *PNAS Nexus*, **2**, pgad150 (2023).
- G. St-Onge, V. Thibeault, A. Allard, L. J. Dubé, and L. Hébert-Dufresne, “[Social confinement and mesoscopic localization of epidemics on networks](#)”, *Phys. Rev. Lett.*, **126**, 098301 (2021).
- G. St-Onge, V. Thibeault, A. Allard, L. J. Dubé, and L. Hébert-Dufresne, “[Master equation analysis of mesoscopic localization in contagion dynamics on higher-order networks](#)”, *Phys. Rev. E*, **103**, 032301 (2021).
- V. Thibeault, G. St-Onge, L. J. Dubé, and P. Desrosiers, “[Threefold way to the dimension reduction of dynamics on networks: An application to synchronization](#)”, *Phys. Rev. Research*, **2**, 043215 (2020).

Preprints

- V. Thibeault, B. Claveau, A. Allard, P. Desrosiers, “[Operator-theoretic approach to the partial integration of randomly coupled phase oscillators](#)”, arXiv, (2026).
- V. Thibeault, B. Claveau, A. Allard, P. Desrosiers, “[Kuramoto meets Kooiman: Constants of motion, symmetries and network motifs](#)”, arXiv, (2025).
- C. Murphy, S. Lizotte, F. Thibault, V. Thibeault, P. Desrosiers, A. Allard, “[On the reconstructability of complex networks from dynamics](#)”, arXiv (2025).
- E. Laurence, C. Murphy, G. St-Onge, X. Roy-Pomerleau, and V. Thibeault, “[Detecting structural perturbations from time series with deep learning](#)”, arXiv (2020).

Invited seminars

- V. Thibeault, “Resilience : from complexity to low-dimensional spaces”, Santa Fe Institute, Santa Fe, New-Mexico (USA), 12/01/2026.
- V. Thibeault, “The low-rank hypothesis of complex systems”, Northwestern University, Chicago, Illinois (USA), 05/06/2025.
- V. Thibeault, “The low-rank hypothesis of complex systems”, University of Oxford Mathematical Institute, Oxford, England, 20/02/2024.
- V. Thibeault, “[The low-rank hypothesis of complex systems](#)”, Network Science Institute, Boston, Massachusetts (USA), 31/01/2024.
- V. Thibeault, “The low-rank hypothesis of complex systems”, Centre Interdisciplinaire de Modélisation Mathématique de l'Université Laval (CIMMUL), Québec, Québec (Canada), 19/04/2024.

Invited talks

- V. Thibeault, “On the dimensionality of dynamics on complex networks: from exact reductions to approximate ones”, SIAM Conference on Uncertainty Quantification, Minneapolis, Minnesota (USA), 23/03/2026.
- V. Thibeault, “Kuramoto meets Koopman: Constants of motion, symmetries, and network motifs”, Northwestern University, Chicago, Illinois (USA), 06/06/2025.

Invited talks and seminars for postdoctoral interview

- Wu Tsai fellowship, Yale, January 8 & January 29, 2026 (talks)
- Omidyar fellowship, Santa Fe Institute, January 12, 2026 (seminar)
- Klarman fellowship, Cornell, December 12, 2025 (talk)
- NITMB fellowship, Chicago & Northwestern, December 4, 2025 (talk)

Contributed talks

- V. Thibeault, “Koopman’s approach to partial integration of the Kuramoto-Sakaguchi model on heterogeneous graphs”, SIAM Conference on Applications of Dynamical Systems, Denver, Colorado (USA), 13/05/2025
- V. Thibeault, “Low-rank nature of connectomes and reducibility of recurrent neural networks”, Montreal AI and Neuroscience (MAIN), Montreal, Québec (Canada), 23/10/2024.
- V. Thibeault, “Revisiting the Kuramoto model on graph: integrals of motion, motifs, and symmetries”, 19th International School and Conference on Network Science (NetSci), Québec, Québec (Canada), 20/06/2024.
- V. Thibeault, “The Low-Rank Hypothesis of Complex Systems – From Empirical and Theoretical Evidence to the Emergence of Higher-Order Interaction”, SIAM Applications in Dynamical Systems, Portland, Oregon (USA), 2023.
- V. Thibeault, “The low-dimension hypothesis implies higher-order interactions in complex systems”, Fourth North-east Regional Conference on Complex Systems (NERCCS), Buffalo, New York (USA), 2022.
- V. Thibeault, “Dimension reduction of high-dimensional dynamics on networks with adaptation”, Networks: A Joint Sunbelt and Netsci Conference, Bloomington, Indiana (USA), 2021.
- V. Thibeault, “Dimension reduction of high-dimensional dynamics on networks with adaptation”, SIAM Applications in Dynamical Systems, Portland, Oregon (USA), 2021.
- V. Thibeault, “Dimension reduction of high-dimensional dynamics on networks with adaptation”, Fourth Northeast Regional Conference on Complex Systems (NERCCS), Buffalo, New York (USA), 2021.
- V. Thibeault, “Réduire la dimension des systèmes complexes: un regard sur l’émergence de la synchronisation”, Centre Interdisciplinaire de Modélisation Mathématique de l’Université Laval, Québec, Québec (Canada), 2020.
- V. Thibeault, B. Klein, B. Case, and F. Normand, “Network analysis of collective motion”, Complex Networks Winter Workshop (CNWW), Québec, Québec (Canada), 2019.
- V. Thibeault, “Predicting synchronization regimes with spectral dimension reduction on graphs”, 14th International School and Conference on Network Science (NetSci), Burlington, Vermont (United States), 2019.
- V. Thibeault, “Predicting synchronization regimes with reduced dimensional dynamics on modular graphs”, Mediterranean School of Complex Networks (MSCx), Salina, Sicily (Italie), 2018.

Thesis

- V. Thibeault, “Approches spectrales et conditions structurelles pour réduire la dimension des systèmes complexes”, PhD thesis, Université Laval (2026).
- V. Thibeault, “[Réduire la dimension des systèmes complexes: un regard sur l’émergence de la synchronisation](#)”, Master’s thesis, Université Laval (2020).

Posters

- B. Claveau, V. Thibeault, A. Allard, P. Desrosiers, “Deciphering large-scale dynamics in complex systems through Koopman operator theory”, Réunion annuelle Sentinelle Nord, Québec, Québec (Canada), 2025.
- V. Thibeault, G. St-Onge, L.J. Dubé and P. Desrosiers, “Threefold way to the dimension reduction of dynamics on networks: an application to synchronization”, 15th International School and Conference on Network Science (NetSci), Rome (Italy), 2020.
- V. Thibeault, X. Roy-Pomerleau, G. St-Onge, J-G. Young, and P. Desrosiers, “Synchronization dynamics on the stochastic block model”, Réunion annuelle Sentinelle Nord, Québec, Québec (Canada), 2018.
- V. Thibeault, X. Roy-Pomerleau, G. St-Onge, J-G. Young, L.J. Dubé and P. Desrosiers, “Synchronization dynamics on the stochastic block model”, 13th International School and Conference on Network Science (NetSci), Paris (France), 2018.
- J. Poirier, N. Fabre, L. Prévost, A. Pappathomas, P. Lamarche, X. Roy-Pomerleau, V. Thibeault, J. Beaudoin-Bertrand, “Generation, detection and usage of attosecond pulses”, 5th International Conference on attosecond physics (ATTO), Ottawa, Ontario (Canada), 2015.

Research impact and outreach

News coverage

J. Gao, "[Intrinsic simplicity of complex systems](#)", Nat. Phys. News & Views, 2024.

T. Gururaj, "[Validating the low-rank hypothesis in complex systems](#)", Phys.org, 2024.

M. De Domenico, "[Complexity Thoughts: Issue #25](#)", 2024.

D. Mackenzie, "[How large a gathering is too large during the coronavirus pandemic?](#)", ScienceNews, 2020.

T. Pueyo, "[Coronavirus: Prevent Seeding and Spreading](#)", Medium, 2020.

Popular science article

G. St-Onge, V. Thibeault, A. Allard, L.J. Dubé, L. Hébert-Dufresne, "[COVID-19: Cancel your next large event, and tell your friends to cancel theirs](#)", Medium, 2020.

Academic service

Student supervision

Benjamin Claveau, Physics Student, 2023-now

Pierre-Luc Larouche, Physics Student, Summer 2021

Peer-review activities

Articles in Nature Physics, Physical Review Letters, Nature Communications, Chaos, Physical Review E, Journal of Physics: Complexity, Nonlinear Dynamics, and European Journal of Applied Mathematics have been reviewed.

Applications for the International School and Conference on Network Science (NetSci) in 2024, 2025, 2026.

Other activities

Presided the session [Data and Koopman Analysis](#) at SIAM Dynamical Systems, 2025.

Awards (in CAD unless stated otherwise)

2026	Wu Tsai Fellowship Wu Tsai Institute, Yale University 90 000\$US/year , Research Funds: 2,000\$US/year, 3 years
2026	Canada Postdoctoral Research Award Conseil de recherches en sciences naturelles et en génie 70 000\$/year, 2 years
2026	Postdoctoral Scholarship of the FRQNT Fonds de recherche Nature et technologies (FRQNT) 55 000\$/year, 2 years
2025	Louis-Berlinguet Rising Star Award for “The low-rank hypothesis of complex systems” Fonds de recherche Nature et technologies (FRQNT) 1500\$
2024	Nature Physics’ Cover Feature Springer Nature
2023	Academic Achievement Scholarship Faculty of Graduate and Postdoctoral Studies, Université Laval 1000\$
2021	Desjardins Excellence Scholarship Caisses Desjardins 2000\$
2021	Publication Award for “Threefold way to the dimension reduction of dynamics on networks: an application to synchronization” CIMMUL 500\$
2019 - 2025	Graduate Research Award Fonds de recherche Nature et technologies (FRQNT) 114 667\$
2020	Master’s Thesis on the Board of Honor Faculty of Graduate and Postdoctoral Studies, Université Laval
2018 - 2019	Graduate Research Award Bourse d’études supérieures du Canada Alexander-Graham-Bell 17 500\$/year
2017 - 2018	Graduate Research Award Fonds de recherche Nature et technologies (FRQNT) 15 000\$/year
2016	Baseball Scholarship Ligue de baseball junior élite du Québec 500\$
2015- 2016	Undergraduate Research Award in Physics and Bioinformatics Faculté des sciences et de génie de l’Université Laval 8 000\$

Volunteering and Involvement

2021 - now	Founder of a philosophy discussion group In the group <i>PEPN</i> , we talk about Montaigne, Camus, Plutarque, Orwell, Nietzsche, Marcus Aurelius, Alain, Pascal, Cioran, Hesse, etc.	Université Laval
2016 - 2020	Co-founder of a mathematical circle for physicists In the mathematical circle <i>Nicole Bourbaki</i> , we talk about logic, combinatorics, graphs, geometry, etc.	Université Laval
2018	Sustainable development Member of a group promoting the reduction of red meat consumption in the cafeterias of the university.	Université Laval
2014 - 2021	Photonic Games Promotion of physics for high school students through optics games.	Université Laval
2015	Welcome Day of Physics	Université Laval
2025 - 2026	Tutor in Differential Calculus and Linear Algebra	Volunteer
2014	Tutor in Integral Calculus	Cégep Sainte-Foy